The HU Collection Part 2

Combinatorial Processes & Invariants

Kevin Hu
Kevin's Website

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Abstract

This collection explores advanced combinatorial processes, algorithmic operations, and state-changes. Mastering the use of invariants, monovariants, and coloring proofs is of paramount importance for elite university admissions; these unorthodox, dynamic problems appear frequently in Oxbridge mathematics and computer science interviews to test a candidate's lateral thinking and rigorous logical structuring.

Introduction to Combinatorial Processes

At its core, a combinatorial process problem follows a simple structural flow:

$$\text{Initial State} + \text{Operation Rules} \longrightarrow \text{Final State}$$

When faced with these dynamic systems, the state space (the number of possible configurations) rapidly becomes overwhelmingly large and brute force becomes barely impossible. Instead, we rely on two primary attack vectors:

1. The Constructive Approach

When given specific operations, the goal is to systematically build the desired end state. This is achieved by:

- **Studying Special Cases:** Looking at small-scale systems ($n = 1, 2, 3$) to find patterns.
- **Induction:** Proving that if a state is reachable for n, it is reachable for $n + 1$.
- **Combining Operations:** Finding a "macro-operation" (a sequence of basic moves) that yields a predictable structural change.

2. State Properties & The State Function, f

$\star$ *This is the most powerful tool in algorithmic combinatorics.*
Instead of tracking the entire chaotic board, we evaluate the system's nature by mapping it to a **State Function**, often denoted as f. This is almost always achieved via:

- **Coloring Proofs** (e.g., checkerboard parity).
- **Weight Assignments / Valuations** (e.g., assigning prime values or algebraic terms to positions).

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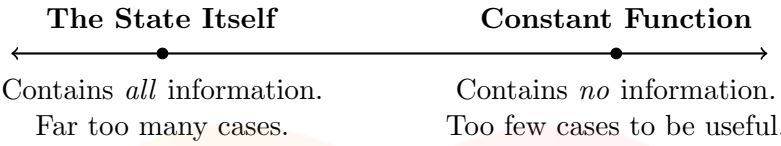
The Goal of Integration

Because the raw problem has too many cases, our objective is to *integrate* the chaos. We wish to find a state function f such that every valid operation induces a simple, predictable change in f. This collapses into two categories:

1. **Invariants:** f remains constant across all operations.
2. **Monovariants (Semi-invariants):** f is strictly non-increasing or strictly non-decreasing. This is vital for proving that a process must eventually terminate.

The Spectrum of f

Choosing the correct state function f is a delicate balancing act on a spectrum:



An effective invariant exists in the middle of this spectrum. It discards the useless noise of the "full state" but retains enough mathematical structure to prove the specific target configuration is unreachable.

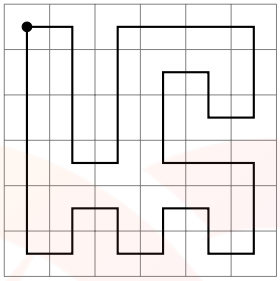
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Problem Set

1. **(MAT 2009)** Given an $n \times n$ grid of squares, where $n > 1$, a *tour* is a path drawn within the grid such that:

   - along its way the path moves, horizontally or vertically, from the centre of one square to the centre of an adjacent square;
   - the path starts and finishes in the same square;
   - the path visits the centre of every other square just once.

   For example, below is a tour drawn in a 6×6 grid of squares which starts and finishes in the top-left square.

   For parts (i)-(iv) it is assumed that n is *even*.

   (i) With the aid of a diagram, show how a tour, which starts and finishes in the top-left square, can be drawn in any $n \times n$ grid.

   (ii) Is a tour still possible if the start/finish point is changed to the centre of a different square? Justify your answer.

   Suppose now that a robot is programmed to move along a tour of an $n \times n$ grid. The robot understands two commands:

   - command R which turns the robot clockwise through a right angle;
   - command F which moves the robot forward to the centre of the next square.

   The robot has a program, a list of commands, which it performs in the given order to complete a tour; say that, in total, command R appears r times in the program and command F appears f times.

   (iii) Initially the robot is in the top-left square pointing to the right. Assuming the first command is an F, what is the value of f? Explain also why $r + 1$ is a multiple of 4.

   (iv) Must the results of part (iii) still hold if the robot starts and finishes at the centre of a different square? Explain your reasoning.

   (v) Show that a tour of an $n \times n$ grid is not possible when n is odd.

2. **(MAT 2020 extension)** During a holiday consisting of exactly n sunny days and n rainy days, Miriam and Adam eat sweets according to the following rules:

   - On any given day, Miriam eats a number of sweets equal to the total number of sunny days that have occurred so far during the holiday (including the current day).
   - Adam eats sweets only on rainy days. If the k-th day of the holiday is rainy, he eats exactly k sweets.

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Prove that by the end of the holiday, Miriam and Adam will have eaten the exact same total number of sweets, regardless of the sequence of the weather.

3. **(BMO1 2024)** Rhian and Jack are playing a game in which initially the number 10^6 is written on a blackboard. If the current number on the board is n, a move consists of choosing two different positive integers a, b such that $n = ab$ and replacing n with $|a - b|$. Rhian starts, then the players make moves alternately. A player loses if they are unable to move.

   Determine, with proof, which player has a winning strategy.

4. Write the numbers $1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{2018}$ on a blackboard. Each time, you choose two different numbers x and y, erase them, and write a new number $x + y + xy$. This operation is repeated until exactly one number remains on the board. Prove that the final remaining number is independent of the order of operations, and find the exact value of this number.

5. **(RMM 2021)** Consider an integer $n \geq 2$ and write the numbers $1, 2, \ldots, n$ down on a board. A move consists in erasing any two numbers a and b, and, for each $c \in \{a+b, |a-b|\}$, writing c down on the board, unless c is already there; if c is already on the board, do nothing. For all integers $n \geq 2$, determine whether it is possible to be left with exactly two numbers on the board after a finite number of moves.

6. **(Russia 2022)** Initially, the pair of numbers $(1, 1)$ is written on a blackboard. For any two real numbers x and y, if one of the pairs $(x, y-1)$ or $(x+y, y+1)$ is already written on the board, you may write the other pair. Similarly, if one of the pairs (x, xy) or $(1/x, y)$ is already on the board, you may write the other pair.

   Prove that the first number in every pair written on the board must always be strictly positive.

7. **(AMO 2014)** Initially, some 1×1 tiles in an $n \times n$ grid are colored. Sally can color any uncolored tile if it shares an edge with at least three already colored tiles. She discovers that by repeating this process, she can eventually color all the tiles in the grid. Prove that the initial number of colored tiles must be greater than $\frac{n^2}{3}$.

8. **(CGMO 2019)** For any sequence of numbers, an allowed operation consists of choosing three adjacent terms, say a, b, c, and replacing them with b, c, a, while keeping all other terms unchanged. Determine all integers $n \geq 3$ such that the sequence $1, 2, \ldots, n$ can be transformed into the reversed sequence $n, n-1, \ldots, 1$ in a finite number of operations.

9. There are 2018 stones placed at various integer coordinates (x, y) on a Cartesian plane. An allowed operation consists of choosing a grid point (a, b) that contains 4 or more stones, removing exactly 4 stones from this point, and placing one stone on each of its four adjacent grid points: $(a, b-1)$, $(a, b+1)$, $(a-1, b)$, and $(a+1, b)$.

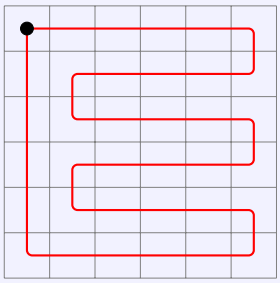
   Prove that after a finite number of operations, the number of stones at every grid point will be at most 3. Furthermore, prove that the final configuration of the stones is independent of the order in which the operations are performed.

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Solutions Appendix

Solution to Question 1

(i) For any even n, a tour can be constructed using a "comb" pattern. The path moves right across the entire top row, weaves down and up the remaining columns, moves left across the bottom row, and returns straight up the left-most column to the start.

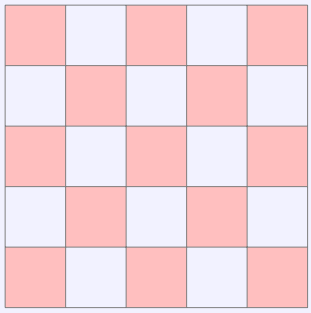


(ii) Yes. A tour forms a continuous, closed loop (*a Hamiltonian cycle*). Because it is a closed loop, the sequence of moves has no distinct beginning or end. Shifting the start/finish point merely cyclically permutes the same sequence of valid operations.

(iii) The robot visits every square in the $n \times n$ grid exactly once and returns to the start, requiring exactly one forward move per square. Thus, $f = n^2$.
Note that the robot initially faces East and ends facing North. If the robot rotates clockwise once more after the journey ends, it will have completed an integer number of full revolutions, hence will have taken a multiple of 4 number of clockwise rotations, hence $r + 1 \equiv 0 \pmod 4$.

(iv) Yes. The total number of steps $f = n^2$ relies solely on the number of squares, which is invariant. However, in the previous part the robot finished its journey a *vertex*, but if it finishes at a straight line section, then we would have already rotated an integer number of full rotations, in which case $r \equiv 0 \pmod 4$.

(v) Color the $n \times n$ grid in a chessboard pattern. Each move F transitions the robot from a square of one color to a square of the opposite color. Any closed tour must therefore contain an even number of steps to return to its starting color. However, a valid tour must visit exactly n^2 squares and requires exactly n^2 steps. If n is odd, n^2 is odd, creating a parity contradiction. Thus, a tour is impossible.



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Solution to Question 2

Let M be the total number of sweets Miriam eats, and A be the total number of sweets Adam eats. We will prove that $A - M = 0$ for all possible arrangements by establishing an invariant for the swapping process.
Consider an arbitrary sequence of weather. Suppose we swap a Rainy day on day k with a Sunny day on day $k + 1$. Let us analyze how this specific transposition affects both totals:

- **Adam's Total (A):** The rainy day shifts from day k to day $k + 1$. Therefore, Adam eats $k + 1$ sweets instead of k sweets for this specific rain event. His total increases by exactly 1. ($\Delta A = +1$)
- **Miriam's Total (M):** Because the sunny day occurs one day earlier (on day k instead of $k + 1$), Miriam will eat exactly 1 extra sweet on day k. For all days before k, the number of sunny days is unchanged. For all days after k, the total accumulation of sunny days remains the same. Her total increases by exactly 1. ($\Delta M = +1$)

Since $\Delta A = \Delta M = 1$, the difference between their totals, $A - M$, remains perfectly invariant under any adjacent swap of (Rain, Sun) to (Sun, Rain). Since any permutation of weather can be achieved purely through adjacent swaps (bubble sort), $A - M$ must be a global constant for all possible sequences.
Because the difference is constant, we can evaluate it for the simplest possible trivial arrangement: all n Rainy days occur first, followed by all n Sunny days.
In this arrangement:

- Adam eats sweets on days 1 through n. His total is the sum of the first n integers:
$$A = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$
- Miriam eats 0 sweets on the first n days (as there has been no sun). On day $n + 1$ she eats 1, on day $n + 2$ she eats 2, and so on until day $2n$ where she eats n. Her total is:
$$M = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

In our base case, $A - M = \frac{n(n+1)}{2} - \frac{n(n+1)}{2} = 0$. Because $A - M$ is invariant across all possible permutations of weather, it must always be true that $A - M = 0$. Thus, Miriam and Adam will always eat the same number of sweets.

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Solution to Question 3

Rhian has a winning strategy.
We can solve this game by analyzing two state functions: a monovariant (to prove termination) and an invariant (to dictate the winning path).
First, we identify the monovariant, or when the process ends. Let n be the current number. A valid move yields $n' = |a - b|$ where $a \neq b$ and $ab = n$. Assuming without loss of generality that $a > b \geq 1$, we know that $a - b < ab = n$. Therefore, the number on the board decreases with every move. Furthermore, because the rules require $a \neq b$, it is impossible to reach 0. The only positive integer with no valid factorization into two distinct positive integers is 1. Thus, the game is finite and must always end at the terminal state $n = 1$.
Now, consider the parity of the number handed to a player:

- Odd State: If a player receives an odd number n, any factorization $n = ab$ guarantees that both a and b must be odd integers. The difference between two odd integers is always even. Thus, a player handed an odd number *must* hand back an even number.
- Even State: If a player receives an even number n, they can always choose the trivial factorization $a = n$ and $b = 1$. The resulting number is $n - 1$. Since n is even, $n - 1$ is strictly odd. Thus, a player handed an even number can *always* choose to hand back an odd number.

Rhian starts the game with $n = 10^6$, which is an even number. Her winning strategy is simply to always hand an odd number to Jack (e.g., by always choosing the $a = n, b = 1$ factorization). Jack, receiving an odd number, is forced to hand an even number back to Rhian.
Because the sequence is strictly decreasing, the game must eventually hit the terminal state $n = 1$. Since 1 is an odd number, and Jack is the *only* player who ever receives odd numbers, it is guaranteed that Jack will eventually be handed the 1, hence Jack will lose.

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Solution to Question 4

We perform basic algebraic manipulation. If we let the newly written number be $z = x + y + xy$, then:

$$z + 1 = x + y + xy + 1 = (x + 1)(y + 1)$$

This reveals a powerful invariant: if we evaluate the quantity $(v + 1)$ for every element v on the board, the overall product of all these quantities remains constant across every operation.

Let P be the invariant product of $(v + 1)$ for all numbers initially on the board:

$$P = (1 + 1)\left(1 + \frac{1}{2}\right)\left(1 + \frac{1}{3}\right)\ldots\left(1 + \frac{1}{2018}\right) = \frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \ldots \frac{2019}{2018} = 2019$$

Let the final remaining number on the board be F. Because the product invariant is preserved throughout the game, it must hold that:

$$F + 1 = P \implies F + 1 = 2019 \implies F = 2018$$

Because F is uniquely determined by the initial state of the board and the invariant of the operation, its final value is 2018, completely independent of the order in which the pairs were chosen.

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Solution to Question 5

Yes, it is always possible to be left with exactly two numbers for all $n \geq 2$.
We will prove that any set $\{1, 2, \ldots, n\}$ can be reduced to the set of even numbers $\{2, 4, 6, \ldots, 2m\}$. Because the operations $a + b$ and $|a - b|$ commute with scalar multiplication, playing the game on $\{2, 4, \ldots, 2m\}$ is isomorphic to playing on $\{1, 2, \ldots, m\}$. Thus, we can continuously halve the state size.
Case 1: $n = 4k$ **or** $n = 4k - 1$
The odd numbers initially on the board are $1, 3, \ldots, 4k - 1$. There are exactly $2k$ odd numbers. We pair them from the outside in:

$$(1, 4k - 1), (3, 4k - 3), \ldots, (2k - 1, 2k + 1)$$

For every pair, the sum is $4k$. The differences are $4k - 2, 4k - 6, \ldots, 2$. Notice that $4k$ and all the differences are even numbers $\leq 4k$. Thus, they are either already present on the board, or they represent $4k$ (which is added if $n = 4k - 1$). After erasing all the odd numbers via these pairs, the board contains exactly the even numbers $\{2, 4, 6, \ldots, 4k\}$. Factoring out 2, this reduces to the case of $n = 2k$.
Case 2: $n = 4k + 1$ **or** $n = 4k + 2$ **(for** $k \geq 2$**)**
First, we apply the operation to the pair $(2k+1, 4)$. The sum is $2k+5$ and the difference is $2k - 3$. Because $k \geq 2$, we have $2k - 3 \geq 1$ and $2k + 5 \leq 4k + 1$. Both results are odd numbers that are already on the board. Thus, erasing $(2k + 1, 4)$ adds no new numbers. The remaining odd numbers are $1, 3, \ldots, 2k - 1$ and $2k + 3, \ldots, 4k + 1$. We pair them from the outside in:

$$(1, 4k + 1), (3, 4k - 1), \ldots, (2k - 1, 2k + 3)$$

For every pair, the sum is exactly $4k + 2$. The differences are exactly $4k, 4k - 4, \ldots, 4$. Notice that the innermost pair generates a difference of 4, which perfectly replaces the 4 we erased in the first step! All generated sums and differences are even numbers $\leq 4k + 2$. After this operation, all odd numbers are erased, and the board contains exactly $\{2, 4, 6, \ldots, 4k + 2\}$. Factoring out 2, this reduces to the case of $n = 2k + 1$.
Base Cases
Case 2 requires $k \geq 2$, meaning we need to verify $n = 5$ and $n = 6$ (which correspond to $k = 1$).

- **For** $n = 5$**:** The board is $\{1, 2, 3, 4, 5\}$. Erase $(2, 3) \to$ sum 5, diff 1 (both already present). Board becomes $\{1, 4, 5\}$. Erase $(1, 5) \to$ sum 6, diff 4 (4 is present). Board becomes $\{4, 6\}$, exactly two numbers.
- **For** $n = 6$**:** The board is $\{1, 2, 3, 4, 5, 6\}$. Erase $(2, 3) \to$ sum 5, diff 1 (both present). Board becomes $\{1, 4, 5, 6\}$. Erase $(1, 5) \to$ sum 6, diff 4 (both present). Board becomes $\{4, 6\}$, exactly two numbers.

By strong induction, the board can always be reduced to two numbers.

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Alternative Solution to Question 5

Yes, it is always possible. We will prove a stronger lemma: *Any set of 3 positive integers* $\{a, b, c\}$ *can be reduced to exactly 2 numbers.* By repeatedly applying this lemma to any three numbers on the board, a set of n numbers will eventually reduce to exactly two.
We will prove this Lemma by Strong Induction on $S = a + b + c$.
First, we define a **macro-operation (*)** that allows us to double any two numbers while leaving the third unchanged:

$$\{a, b, c\} \xrightarrow{\text{operate on } a,b} \{a + b, |a - b|, c\} \xrightarrow{\text{operate on } a+b,|a-b|} \{2a, 2b, c\}$$

Because the game operations are linear, if all numbers on the board share a common divisor d, the game is isomorphic to playing with the numbers divided by d.
Case 1: $\gcd(a, b, c) = d > 1$
We factor out d. The state reduces to $\{a/d, b/d, c/d\}$. The new sum is $S/d < S$. By strong induction, this reduces to 2 numbers.
Case 2: $\gcd(a, b, c) = 1$
Case 2.1: At least one number is even.
Without loss of generality (WLOG), let c be even. Apply macro (*) to a and b to get $\{2a, 2b, c\}$. Now, all three numbers are even. Factoring out 2, the state scales down to $\{a, b, c/2\}$. The new sum is $a + b + c/2 < S$. By induction, this reduces to 2 numbers.
Case 2.2: All three numbers are odd.
WLOG, let $a > b$. We perform the following sequence:

1. Operate on a and b: The board becomes $\{a + b, a - b, c\}$. Note that both $a + b$ and $a - b$ are even, while c remains odd.
2. Apply macro (*) to $a - b$ and c: The board becomes $\{a + b, 2(a - b), 2c\}$.
3. Apply macro (*) to $a + b$ and $2c$: The board becomes $\{2(a + b), 2(a - b), 4c\}$.

Since a and b are odd, both $a + b$ and $a - b$ are even integers. Therefore, $2(a + b)$ and $2(a - b)$ are multiples of 4. Now, all three numbers on the board are multiples of 4. Factoring out 4, the state scales down to:

$$\left\{\frac{a + b}{2}, \frac{a - b}{2}, c\right\}$$

The new sum is $\frac{a+b}{2} + \frac{a-b}{2} + c = a + c$. Because $b > 0$, we have $a + c < a + b + c = S$. By strong induction, this reduces to 2 numbers.
Thus, for all cases, the set inevitably collapses, meaning any $n \geq 2$ starting numbers can be reduced to exactly two.

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Solution to Question 6

We define the state function $f(x, y) = y^2 - 4x$. We will analyze how this function behaves under the two allowed operations.
Evaluate the state function for both pairs:

$$\begin{aligned} f(x+y, y+1) &= (y+1)^2 - 4(x+y) \\ &= y^2 + 2y + 1 - 4x - 4y \\ &= (y-1)^2 - 4x \\ &= f(x, y-1) \end{aligned}$$

Thus, the value of $f(x, y)$ is invariant under the first operation.
Evaluate the relationship between the two pairs:

$$f(x, xy) = (xy)^2 - 4x = x^2y^2 - 4x$$

$$f\left(\frac{1}{x}, y\right) = y^2 - \frac{4}{x}$$

Notice that $f(x, xy) = x^2 \cdot f\left(\frac{1}{x}, y\right)$. Because $x^2 > 0$ (assuming $x \neq 0$), multiplying by a positive square does not change the sign. Thus, the *sign* of $f(x, y)$ is invariant under the second operation.
The initial state of the board is the pair $(1, 1)$.

$$f(1, 1) = 1^2 - 4(1) = -3 < 0$$

Since the operations preserve either the exact value or the sign of f, every pair (x, y) written on the board must satisfy $f(x, y) < 0$.
Therefore, for any valid pair on the board:

$$y^2 - 4x < 0 \implies x > \frac{y^2}{4} \geq 0$$

This requires $x > 0$, hence first number in any pair is always positive.

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Solution to Question 7

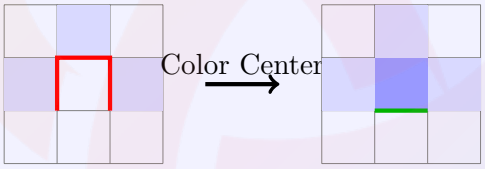
We will solve this using a geometric monovariant.
To eliminate edge-case anomalies at the borders of the grid, we embed the $n \times n$ grid inside a larger $(n+2) \times (n+2)$ grid. The outer border of this new grid consists entirely of uncolored (white) tiles, which can never be colored.
Let S be the total number of shared edges between a colored tile and an uncolored tile (effectively the "perimeter" of the colored regions). Let k be the initial number of colored tiles. Since each colored tile has exactly 4 edges, the maximum possible initial perimeter is when all k tiles are completely isolated. Thus:

$$S_{initial} \leq 4k$$

When Sally colors a new tile, that tile must have had at least 3 colored neighbors.

- If it had exactly 3 colored neighbors, coloring it absorbs (removes) 3 colored-uncolored edges, and introduces 1 new colored-uncolored edge. The net change in S is -2.
- If it had exactly 4 colored neighbors, coloring it absorbs 4 colored-uncolored edges, and introduces 0 new ones. The net change in S is -4.

In all valid cases, every operation decreases the total perimeter S by *at least* 2.



Before: 3 shared edges After: 1 new edge (Net: -2)

The process stops when all n^2 tiles in the $n \times n$ grid are colored. At this point, the colored region is a solid $n \times n$ square bounded by the virtual uncolored ring. The final perimeter is exactly the boundary of this square:

$$S_{final} = 4n$$

Since we started with k colored tiles and ended with n^2 colored tiles, exactly $n^2 - k$ operations were performed. Because each operation decreases S by at least 2, we have:

$$S_{initial} - S_{final} \geq 2(n^2 - k)$$

Substitute our known bounds ($S_{initial} \leq 4k$ and $S_{final} = 4n$):

$$4k - 4n \geq 2n^2 - 2k$$

$$k \geq \frac{n^2 + 2n}{3}$$

Since $n \geq 1$ (a grid must exist), it follows that $\frac{2n}{3} > 0$. Therefore:

$$k \geq \frac{n^2}{3} + \frac{2n}{3} > \frac{n^2}{3}$$

Thus, more than $\frac{n^2}{3}$ tiles must be initially colored.

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Solution to Question 8

The sequence can be reversed if and only if $n \equiv 0 \pmod 4$ **or** $n \equiv 1 \pmod 4$.
The operation $(a, b, c) \to (b, c, a)$ is a cyclic shift of 3 elements. In combinatorics, this is equivalent to two adjacent transpositions (swapping a, b then a, c). Because the number of swaps is even, the operation preserves the parity of the total number of inversions in the sequence.
The initial sequence $1, 2, \ldots, n$ has 0 inversions (an even number). The target sequence $n, n-1, \ldots, 1$ has exactly $\frac{n(n-1)}{2}$ inversions. For the target to be reachable, $\frac{n(n-1)}{2}$ must be even, which implies $n(n-1)$ must be divisible by 4. This only occurs when $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$.
Notice that applying the allowed operation twice on the same three elements yields $(a, b, c) \to (b, c, a) \to (c, a, b)$. This macro-operation effectively moves c two positions to the left (or a two positions to the right) without altering the relative order of the other elements. By repeating this, we can move any element left or right by any even number of positions without disturbing the rest of the sequence.

- $n = 4$**:** Start with $(1, 2, 3, 4)$. Move 4 left by two positions to get $(1, 4, 2, 3)$. Move 1 right by two positions to get $(4, 2, 1, 3)$. Move 3 left by two positions to get $(4, 3, 2, 1)$.
- $n = 5$**:** Start with $(1, 2, 3, 4, 5)$. Move 5 left by four positions (two jumps of two) to get $(5, 1, 2, 3, 4)$. Then, apply the $n = 4$ algorithm to the remaining block to get $(5, 4, 3, 2, 1)$.

Assume the reversal is possible for $n - 4$. Consider a sequence of length n. We partition it into two blocks: block $A = (1, \ldots, n-4)$ and block $B = (n-3, n-2, n-1, n)$.
Because block A has an even length ($n-4$ is even since $n \equiv 0, 1 \pmod 4$), we can use our leapfrog macro-operation to move every element of block B to the left side of block A by jumping 2 elements at a time. This shifts block B entirely to the front without altering the internal order of A or B. The sequence is now $(n-3, n-2, n-1, n, 1, 2, \ldots, n-4)$.
Finally, we apply the base case $n = 4$ algorithm to reverse the first block, and the inductive hypothesis algorithm to reverse the second block, yielding exactly $(n, n-1, \ldots, 1)$.
Thus, by mathematical induction, all $n \equiv 0, 1 \pmod 4$ are achievable.

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Solution to Question 9

Termination: Let $n = 2018$ and denote the coordinates of the stones as (x_k, y_k). Define two state functions: $f = \sum_{k=1}^{n}(x_k^2 + y_k^2)$ and $g = \sum_{i<j}(|x_i - x_j| + |y_i - y_j|)$.
An operation at (a, b) moves 4 stones to $(a \pm 1, b)$ and $(a, b \pm 1)$. Expanding the sum of squares yields $\Delta f = 4$. After K operations, $f_K = f_0 + 4K$.
To evaluate Δg, we partition the pairwise distances. The internal Manhattan distance between the 4 moving stones increases from 0 to 12 (two opposite pairs of distance 2, and four adjacent pairs of distance 2). For any stationary stone at (x, y), its total distance to the 4 moving stones changes from $4(|x - a| + |y - b|)$ to:

$$(|x - a - 1| + |x - a + 1| + 2|x - a|) + (|y - b - 1| + |y - b + 1| + 2|y - b|)$$

By the triangle inequality ($|u - 1| + |u + 1| \geq 2|u|$), this new distance is at least $4|x - a| + 4|y - b|$. Thus, cross-distances cannot decrease. Therefore, $\Delta g \geq 12$, yielding $g_K \geq g_0 + 12K$.
As noted in the algebraic bounding method, we relate g to f via the Cauchy-Schwarz inequality. For the x-coordinates:

$$\left(\sum_{i<j}|x_i - x_j|\right)^2 \leq \frac{n(n-1)}{2}\sum_{i<j}(x_i - x_j)^2 = \frac{n(n-1)}{2}\left[n\sum_{i=1}^{n}x_i^2 - \left(\sum_{i=1}^{n}x_i\right)^2\right]$$

$$\leq \frac{n^2(n-1)}{2}\sum_{i=1}^{n}x_i^2$$

Applying this symmetry to the y-coordinates implies $g_K^2 \leq C \cdot f_K$, where C is a constant dependent only on n. Substituting our sequence bounds gives $(g_0 + 12K)^2 \leq C(f_0 + 4K)$. A quadratic function in K cannot be bounded by a linear function for all K. Thus, the total number of operations K is finite.
Uniqueness: Let Φ be the initial state, and let T_k denote the operation of firing grid point k. Operations are commutative ($T_aT_b(\Phi) = T_bT_a(\Phi)$) because firing a only adds stones to its neighbors, which never invalidates a pending valid move at b.
Suppose two maximal finite sequences lead to termination. If the first sequence begins with T_{a_1}, then a_1 must eventually be fired in the second sequence. Let T_{b_i} be the first occurrence of firing a_1 in the second sequence.
Because operations commute, we can slide T_{b_i} to the very front of the second sequence without changing the final state:

$$T_{b_m} \ldots T_{b_i} \ldots T_{b_1}(\Phi) = T_{b_m} \ldots T_{a_1} \ldots T_{b_1}(\Phi) = T_{b_m} \ldots T_{b_1}T_{a_1}(\Phi)$$

We can now cancel T_{a_1} from the beginning of both sequences. By repeatedly sliding and canceling every subsequent operation, we show both sequences consist of the exact same multiset of operations, proving the final arrangement is unique.

Kevinvariant
kevinvariant08.github.io